Lesson 1

AIR-CONDITIONING & MECHANICAL VENTILATION

Official Open

MODULE LECTURER

David Teo 轩维

Tel: 98828530

E-mail: tsh52@np.edu.sg

MODULE INTRODUCTION

Official Open

Syllabus

* Module Code : NP-DACMV
* Level : 2
* Pre-requisites : Basic Air-conditioning
* **Contact Hours**
* Lecture/Tutorial : 4 hours
  Practical

Official Open

Assessment Weightage

CT: Common Test (individual) : 20 %

WTEST: Written Test(individual) : 20%

ASG – Assignment (Group) : 40%

CLWK : Regular Class Work (Individual) : 20%

Syllabus

Module Name: Air-Conditioning & Mechanical Ventilation				
Module ID: NP-DACMV/NP-012406				
Time: 6.00 to 10.00pm				
Module Leader: David Teo		Email: TEO_Seow_Hian@np.edu.sg		
Lesson	**Date (Tue/Thurs)**	**Venue**	**Topic**	**Lecturer**
1	23/4/2026 (Thu)	**LT 58B**	Types of Air-conditioning Systems	David Teo
2	30/4/2026 (Thu)	MS Teams	Air-conditioning Equipment	David Teo
3	7/5/2026 (Thu)	MS Teams	Duct Systems & Sizing	David Teo
4	14/5/2026 (Thu)	**LT 58B**	Duct Systems & Sizing	David Teo
5	Group A 21/5/2026 (Thu) Group B 22/5/2026 (Fri)	34-03-02	Practical 1	David Teo / Gan Poh Tin
6	28/5/2026 (Thu)	MS Teams	Pipe Systems & Sizing	David Teo
7	4/6/2026 (Thu)	**LT 58B**	Common Test	David Teo
8	11/6/2026 (Thu)	**LT 58B**	Pipe Systems & Sizing	David Teo
		Vacation Break (15 Jun 2026 to 28 Jun 2026)		

Syllabus

	(Thu)			
		Vacation Break (15 Jun 2026 to 28 Jun 2026)		
9	2/7/2026 (Thu)	MS Teams	Air Systems, Indoor and Outdoor Design Conditions System	David Teo
10	9/7/2026 (Thu)	LT 58B	Envelope Thermal Performance	David Teo
11	Group A 16/7/2026 (Thu) Group B 17/7/2026 (Fri)	34-02-0006	Practicals 2 & 3	David Teo/Gan Poh Tin
12	23/7/2026 (Thu)	LT 58B	Cooling Load Estimation	David Teo
13	30/7/2026 (Thu)	MS Teams	System Psychrometrics	David Teo
14	6/8/2026 (Thu)	MS Teams	System Psychrometrics	David Teo
15	13/8/2026 (Thu)	LT 58B	Test 2	David Teo

References

1	Principles of Heating, Ventilating and Air-conditioning - Ronald H Howell, Harry J Sauer, Jr, William J Coad - ASHRAE
2	Principles of Refrigeration - R J Dossat
3	American Society of Heating, Refrigerating & Air-conditioning Engineers, Inc (ASHRAE) Handbooks - Fundamentals - HVAC Equipment & Systems - HVAC Applications - Refrigeration

References

4	Carrier Handbook on Air-conditioning
5	Singapore Standard SS 553: 2016 – CP on Mechanical Ventilation and Air-conditioning in Buildings
6	Code of Practice on Control of Legionella Bacteria in Cooling Towers (National Environment Agency)

Practical Sessions

1	Carrier E-20 Software	
2	Performance of a Central Station Air Handling Unit	
3	Performance of Water-cooled Screw Chiller	